

NeuralShield AI

Real-time Scam and Fraud Detection

Across Text -- Images -- URLs -- QR Codes -- Voice

Detailed Product and Business Overview | 2025

Executive Summary

Neural Shield AI is a production-ready, multi-modal fraud detection platform designed to protect individuals, businesses, and platforms from the full spectrum of digital scams. The platform analyzes any piece of digital content -- text messages, images, URLs, QR codes, email, UPI payment identifiers, and phone numbers -- through a deterministic trust engine, returning a structured verdict in under two seconds.

The platform's core architecture separates scoring from explanation. A deterministic pipeline of rules, threat intelligence collectors, and community reputation signals produces the final risk score. An LLM (Claude by Anthropic) then explains that score in plain language -- but never generates the numbers. This separation makes Neural Shield AI auditable, reproducible, and trustworthy in ways that pure AI-based fraud detection cannot be.

As of 2025, the platform operates in production across three channels: a web application for end users and businesses, a Chrome browser extension for real-time link and page protection, and a REST API for enterprise integration. All three channels share the same underlying trust engine, ensuring consistent verdicts regardless of how the platform is accessed.

Key Highlights

Dimension	Detail
Stage	Live in production -- web app, Chrome extension, REST API all active
Engine Version	Trust Engine v2.1 -- deterministic scoring, 11 TI collectors, community reputation
Modalities Covered	Text, URL, Email, Phone, UPI/Payment, Screenshot (OCR), QR Code
Processing Time	Under 2 seconds for most scans; under 4 seconds for full TI + AI explanation
Cost Per Scan	Under USD 0.005 for text/URL; under USD 0.02 for image+OCR+TI
Primary Markets	India (initial), Southeast Asia, Global English-speaking markets
Revenue Model	Freemium (web/extension) + API subscription (B2B) + Enterprise licensing
Target Customers	Individuals, Fintechs, Telecoms, E-commerce, Government/Regulatory bodies

The Investment Case

Fraud detection is not a new market -- it is a rapidly expanding one driven by the industrialization of scam operations. Organized criminal networks now operate at scale with toolkits that generate thousands of personalized phishing messages per hour, register dozens of lookalike domains per day, and synthesize voice calls impersonating government officials. Consumer-facing fraud detection at this scale does not yet exist as a standalone product.

Neural Shield AI targets the gap between enterprise security tools (too expensive and complex for individuals and small fintechs) and basic spam filters (too blunt to catch context-aware attacks). The platform has a defensible technical moat -- the signal weight matrix, collector architecture, and community reputation layer take significant time to tune -- and expands naturally with each new modality added. Every collector plugged in increases detection breadth for all existing customers without code changes.

Company Overview

Neural Shield AI was founded with a single premise: that ordinary people should have access to the same fraud detection intelligence that banks and security firms use internally, delivered at a price and speed suitable for everyday use. The company develops detection technology for digital content and positions itself at the intersection of cybersecurity, consumer protection, and AI-augmented analysis.

Mission

To make it effortless for anyone -- regardless of technical skill -- to verify before they trust, in every digital interaction.

Core Principles

- **Determinism over opaqueness:** Every score is the sum of auditable signals with documented weights. No black-box AI decides whether something is a scam.
- **Fail-open architecture:** A collector failure lowers confidence but never crashes a scan. Availability is never sacrificed for coverage.
- **Privacy by design:** The backend holds no service-role database key. All user data access is governed by Row Level Security enforced at the database layer.
- **Modality agnosticism:** The underlying signal accumulation model is the same for URLs, phone numbers, UPI IDs, and text. Adding a new entity type is additive, not architectural.
- **Honest AI:** LLMs explain verdicts -- they do not produce them. This keeps the system auditable and prevents hallucinated risk scores.

Current Status

As of June 2025, the platform is live in production with paying users on the Pro tier. The trust engine has processed thousands of scans across all modalities. The Chrome extension is published and active. The REST API accepts both JWT (web users) and API key (enterprise/developer) authentication. All backend infrastructure runs on Vercel serverless functions backed by Supabase PostgreSQL with full Row Level Security.

Problem Statement

The Scale of Digital Fraud

Digital fraud has crossed a threshold where it can no longer be addressed with awareness campaigns and reactive blacklists. In India alone, the National Crime Records Bureau reported over 7,000 cyber fraud incidents every single day in 2023. The Global Anti-Scam Alliance estimates that global scam-related losses exceeded USD 1.02 trillion in 2023 -- a figure that surpasses the GDP of most countries. These are not isolated incidents; they are the output of industrialized criminal operations running at software scale.

The attack surface has expanded dramatically in the last three years. Scammers in 2020 primarily operated through bulk SMS phishing. Today they combine personalized WhatsApp messages, AI-generated voice calls impersonating bank officials, fake payment screenshots used as proof of transfer, QR codes embedded in physical flyers at local markets, and URLs registered hours before a campaign launches -- expiring before blocklists can catch them. Each vector requires different detection methods, and victims encounter multiple vectors simultaneously.

Attack Vector Breakdown

Attack Vector	Method	Why Hard to Detect
SMS/WhatsApp phishing	Personalized messages with urgent language + embedded links	Context-aware; passes keyword filters; links freshly registered
Fake payment screenshots	Altered UPI success screen sent as 'proof' of payment	Images bypass all text analysis; OCR rarely applied by victims
QR code scams	Physical flyers or forwarded images with encoded URLs or UPI IDs	Destination hidden until scan; widely trusted format
Voice call impersonation	AI-synthesized voice claims to be from SBI, RBI, or IT dept.	Real-time; no digital artifact to scan; high psychological pressure
UPI collect requests	Scammer sends collect request (not a payment link)	Looks identical to receiving money; many users approve without reading
Job/investment fraud	Fake job offers or investment groups on Telegram/WhatsApp	Gradual trust-building; no single message is obviously fraudulent
Lookalike domains	hdfc-loan.xyz registers same day, used in phishing campaign	Domain age under 24h; TLS certificate makes it look legitimate
KYC fraud	Message claims KYC expiry; requests Aadhaar/PAN details or OTP	Official-looking language; exploits fear of service disruption

Who Gets Hurt Most

The demographic most severely affected by digital fraud is not the least careful -- it is the least equipped. Elderly users targeted by authority impersonation (fake bank calls, TRAI SIM suspension threats) frequently lose life savings in a single interaction. Rural populations encounter QR code scams at local markets and kirana stores, where printed QR codes have been replaced by scammers overnight. Small and medium businesses lose money to fake invoice URLs and fraudulent UPI collect requests from

impersonated suppliers. Financial platforms absorb fraud losses through chargebacks and disputes when their customers are deceived before a transaction reaches the bank's fraud detection layer.

The economic damage ripples outward. When a user is defrauded via a fintech platform's UPI integration, the platform loses the user's trust even if it was not at fault. Telecom providers face regulatory pressure when their networks carry scam SMS at scale. E-commerce platforms lose seller trust when fraudulent payment screenshots are accepted as proof of payment. There is systemic value in solving this problem -- not just individual value.

Why Existing Solutions Fall Short

The fraud detection landscape today is fragmented by modality and by tier. Large enterprises deploy SIEM systems costing millions of dollars annually, calibrated for network-layer threats rather than content-layer scams. Consumer-facing tools are either modality-specific (URL scanners, email spam filters) or reactive (blocklists that lag campaigns by hours or days). No single product handles text, images, URLs, QR codes, phone numbers, and payment identifiers in a unified pipeline.

- **Spam filters:** Block known keywords but miss context. 'Your KYC has expired' with a freshly registered domain bypasses every keyword list.
- **Reverse image search:** Requires deliberate user effort, finds only indexed images, and fails on screenshots of fraudulent UIs that have never been indexed.
- **URL scanners:** Miss domains registered hours before a campaign. VirusTotal adds value for known-bad domains but has zero coverage on day-zero attacks.
- **Phone number lookups:** Typically return carrier information only. Scam pattern detection requires prefix tables, digit sequence analysis, and community reports.
- **Static blocklists:** By definition reactive. A scam campaign that runs for 4 hours and then migrates to a new domain is invisible to any blocklist.
- **AI-only detection:** LLMs hallucinate risk scores, produce inconsistent verdicts for the same input, and cannot cite evidence for their conclusions -- making them unusable for financial and regulatory contexts.

Our Solution

Neural Shield AI solves the fragmentation problem with a unified multi-modal detection pipeline. Any digital content -- regardless of type -- enters the same orchestration layer, is classified and normalized, routed to the appropriate analysis sub-pipelines, and emerges with a structured verdict: a numeric risk score, a risk level label, a confidence score, a list of evidence signals, and a plain-language explanation.

The platform is not a product bolted together from third-party APIs. The trust engine is a custom-built, versioned scoring system where every weight, threshold, and collector is documented, testable, and reproducible. ENGINE_VERSION is embedded in every scan result, so a verdict from six months ago can be replicated exactly with the same engine version.

The Core Insight

Scams are multi-signal events. A fraudulent SMS contains urgency language, a freshly registered domain, a UPI ID with a suspicious handle, and a phone number starting with a premium-rate prefix -- simultaneously. Analyzing signals in aggregate catches what single-modality tools miss.

How Verdicts Are Structured

Output Field	Type	Description
scamProbability	0-100 integer	Percentage likelihood this content is a scam (from risk score)
trustScore	0-100 integer	Inverse of scam probability; higher is safer
riskScore	0-100 float	Raw accumulated signal score before clamping
riskLevel	Enum	safe / low / medium / high / critical
confidence	0-1 float	How much signal coverage supported this verdict
signals[]	Array	Each fired signal with id, weight, confidence, source, evidence
flags[]	Array	Human-readable flags for the UI (derived from signals)
summary	String	Plain-language explanation (generated by Claude)
recommendation	String	Actionable advice for the user
scamType	String	Category of scam detected (e.g. kyc_fraud, phishing, investment_fraud)
engineVersion	String	Versioned trust engine id (e.g. trust-engine@2.1.3)
processingTimeMs	Integer	End-to-end latency in milliseconds

Risk Level Bands

Label	Risk Score Range	Trust Score	Recommended Action
Safe	0 - 4	96-100	No action needed; proceed normally

Low Risk	5 - 19	81-95	Be aware; verify sender if requesting money
Medium	20 - 49	51-80	Suspicious -- do not share credentials or pay
High	50 - 79	21-50	Likely scam -- avoid interaction entirely
Critical	80+	0-20	Confirmed or near-confirmed scam -- block and report

Product Features -- In Depth

Each analysis modality is a distinct sub-pipeline with its own normalization, rules, collectors, and signal set. They share the same risk accumulation model and confidence calculation, ensuring consistent output format regardless of input type.

5.1 Text / SMS / WhatsApp Analysis

The text pipeline is the most broadly applicable. Any message -- a WhatsApp forward, an SMS alert, a Telegram post -- can be submitted. The engine first runs the full content rules set (10 pattern families) over the raw text. It then extracts embedded entities: URLs, UPI IDs, phone numbers, and email addresses. Each extracted entity is run through its own sub-pipeline in parallel. The final signal set includes signals from both the text analysis and all embedded entity analyses.

- **Credential request detection:** Regex over OTP, PIN, CVV, password, MPIN, UPI PIN variants. Confidence 0.9 -- the highest content rule confidence.
- **KYC/Aadhaar pressure:** Detects requests referencing KYC, re-KYC, Aadhaar, PAN in contexts implying urgency or account suspension.
- **Urgency and threat language:** Catches 'blocked', 'suspended', 'deactivated', 'expires within 24 hours', 'act now', 'last warning' -- the pressure tactics common to all scam categories.
- **Lottery and prize bait:** Covers lottery, prize, won, winner, KBC, lucky draw, reward, gift, cashback patterns.
- **Job fraud:** Detects work-from-home offers, part-time job with daily income claims, registration fees, joining fees, and security deposit requests.
- **Loan fraud:** Unsolicited loan offers with processing fee requests.
- **Investment fraud:** 'Double your money', 'guaranteed returns', 'risk-free', percentage daily/weekly/monthly profit claims, trading signals.
- **Off-platform migration:** Attempts to move the conversation to WhatsApp, Telegram, or other unmonitored channels.
- **APK / malware delivery:** Requests to install an APK, enable unknown sources, or download an attachment.
- **Brand impersonation:** Detection of 21 major Indian and global brands (SBI, HDFC, ICICI, Axis, Kotak, PayTM, PhonePe, Google Pay, NPCI, TRAI, Income Tax, EPFO, UIDAI, IRCTC, Amazon, Flipkart, Netflix, Microsoft, Apple, PayPal, RBI) used in scam context.

5.2 URL Scanning

The URL pipeline runs 11 collectors in parallel under a 4-second global budget. Each collector is independently fail-open: a timeout or API error returns an empty signal set and reduces confidence but does not block the scan. Results from each collector are cached per-URL (TTL varies by source: 1 hour for live threat feeds, 4 hours for domain age data) to avoid redundant API calls on repeated scans of the same URL.

Collector	Signal IDs	Coverage
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RDAP (domain age)	domain.age_lt_7d, age_lt_30d, age_lt_90d, age_lt_1y, age_gt_2y, age_gt_5y	Domain registration date
TLS certificate	infra.tls_self_signed, tls_cn_mismatch, infra.no_tls	Certificate validity and hostname match
Redirect expansion	infra.redirect_chain_long	Unrolls up to 10 hops; catches shortener abuse
Google Safe Browsing	ti.gsb.malware, ti.gsb.social_engineering	Google's malware + phishing database
URLHaus	ti.urlhaus.malware	Malware distribution URLs
PhishTank	ti.phishtank.verified	Community-verified phishing URLs
OpenPhish	ti.openphish.match	Real-time phishing feed
VirusTotal v3	ti.virustotal.ratio_high, ratio_low	Aggregated AV engine verdicts
Spamhaus DBL/ZEN	ti.spamhaus.dbl, ti.spamhaus.zen	Domain and IP blocklists
AbuseIPDB	ti.abuseipdb.high, ti.abuseipdb.low	Hosting IP abuse reports
DNS checks	domain.low_ttl, domain.sinkholed, domain.no_mx_yet_emails	TTL anomaly, sinkhole IP detection

Structural URL heuristics run before network collectors and add signals for raw IP addresses in the host position, punycode homoglyph domains, brand names placed in subdomains, suspicious TLDs (.xyz, .top, .win, .click), excessive subdomain nesting, '@' symbols in the URL path, and HTTP (no TLS) connections.

5.3 Email Analysis

Email analysis handles a composite blob: sender address, subject line, and body are submitted together. The engine normalizes the sender address to split local and domain parts, runs the full content rules set over the composed text, and dispatches the sender domain through the TI pipeline (RDAP, DNS, Spamhaus, AbuseIPDB). Two email-specific identity rules add high-value signals:

- **Sender domain mismatch:** If the email body mentions a brand (HDFC, SBI, etc.) and the sender's domain does not match that brand's canonical sending domain, the signal fires. For example, a message claiming to be from HDFC Bank sent from hdfcalert.com fires sender_domain_mismatch with 0.8 confidence.
- **Free email impersonation:** If the sender is from Gmail, Yahoo, Outlook, Hotmail, Rediffmail, Ymail, or Proton.me and the body mentions a corporate brand, the free_email_for_company signal fires. This catches 'sbi.support@gmail.com' impersonation patterns.
- **SPF/DKIM/DMARC pass:** When authentication headers confirm the sender, a negative signal (-12 weight) provides a modest trust boost -- reducing false positives on legitimate bank communications.
- **Missing MX record:** If the sender domain has no MX record but is used in an email context, domain.no_mx_yet_emails fires -- indicating a domain registered purely for phishing, not as a real email host.

5.4 UPI / Payment ID Analysis

UPI analysis covers both plain VPA format (handle@psp) and full intent URI format (upi://pay?pa=handle@psp&pn;=PayeeName&am;=Amount). The engine extracts the VPA from either format, splits into handle and PSP components, and runs four independent rule checks:

- **Unknown PSP:** The PSP (part after '@') is checked against a curated allowlist of 35 known legitimate payment service providers (oksbi, okhdfcbank, okicici, paytm, ybl, ibl, etc.). A PSP not on this list fires pay.upi_unknown_psp.
- **Brand impersonation in handle:** If the handle contains a BRAND_RE match AND a scam keyword (refund, kyc, verify, support, help, care, prize, bonus), pay.upi_brand_impersonation fires. Example: 'sbi.refund@upi'.
- **Suspicious handle keyword:** Even without a brand name, the handle is checked against 16 scam-associated keywords (prize, winner, lottery, lucky, reward, refund, cashback, kyc, verify, verification, support, helpdesk, helpline, care, bonus, free, gift, claim). Catches 'prize.winner123@hdfcbank' which has a known PSP but a scam handle.
- **Intent URI name mismatch:** When a UPI intent URI provides a payee name (pn= parameter) claiming a brand, but the VPA's PSP does not match that brand's official PSP list, pay.upi_intent_name_mismatch fires. Example: pn='SBI Bank' but pa='random123@paytm'.
- **Collect request context:** When a UPI ID appears alongside text mentioning 'collect request', 'approve', 'accept payment', 'pending collect', etc., pay.collect_request_unsolicited fires -- alerting users that this is a money pull, not a payment.

5.5 Phone Number Analysis

Phone analysis normalizes all numbers to India-first E.164. Bare 10-digit numbers are assumed to be Indian mobile and prefixed with +91. The engine then runs four rule checks:

- **Invalid Indian mobile format:** TRAI allocates Indian mobile numbers starting with digits 6, 7, 8, or 9 only. A number with +91 prefix whose local digits start with 0-5 is not a valid Indian mobile -- it may be a spoofed foreign number or a fabricated identifier. Weight 30, Tier 1, confidence 0.95.
- **Premium-rate prefix:** Prefixes 900x (premium rate voice) and 140x/141x (VOIP/telemarketing, per TRAI DIP registry) are flagged. Weight 35, Tier 1 -- the highest phone signal weight.
- **Suspicious digit sequence:** Numbers with all identical digits (9999999999) or known sequential patterns (1234567890, 9876543210, 0987654321) are flagged as test/fake numbers. Weight 20, Tier 2.
- **Non-Indian claiming Indian bank:** If the phone number is not Indian (does not start with +91 or is not 12 digits with 91 prefix) but the surrounding text mentions an Indian bank or authority brand, phone.intl_claiming_indian_bank fires. Catches foreign scam call centers impersonating Indian banks.

5.6 Screenshot / Image Analysis

Screenshot analysis uses Tesseract OCR (via tesseract.js) to extract text from uploaded images. The extracted text then runs through the full text pipeline including entity extraction. This allows Neural Shield AI to catch:

- Fake UPI payment success screenshots sent as 'proof' of payment in buy/sell transactions
- Fraudulent banking app screens showing fabricated balances or transaction histories
- Fake employment offer letters with scam patterns in the body text

- Screenshots of scam WhatsApp messages forwarded without the original message context
- Fake lottery or prize notification images with embedded claim links

OCR quality depends on image resolution and clarity. The engine returns a 422 error with a user-friendly message when extracted text is too short (under 3 non-whitespace characters) to analyze reliably, rather than returning a meaningless safe verdict.

5.7 QR Code Decoding

QR code analysis uses jsqr to decode the payload from an uploaded image. The decoded content is then auto-detected by type and routed to the appropriate sub-pipeline. A URL payload routes to the URL scanner. A UPI intent URI routes to the UPI analyzer. Raw text routes to the text pipeline. An `infra.via_qr_code` signal (weight 5, Tier 3) is always injected when content arrives via QR, flagging that the link destination was hidden before scanning -- a common pattern in physical-world scam campaigns.

5.8 Chrome Browser Extension

The Chrome extension (Manifest V3) provides real-time protection while browsing. It integrates directly with the Neural Shield AI backend using the same authentication and scoring as the web app. Key capabilities:

- Analyzes the current tab's URL on demand via popup
- Shows inline verdict (risk level, trust score, detected flags) without leaving the page
- Offline rule-based checks for zero-latency feedback on obviously suspicious patterns before the network call completes
- Stores analysis history in `chrome.storage.local` for session-level review
- Supabase Auth integration -- users sign in once and the extension uses the same Pro account as the web app
- Batch analyze endpoint support for scanning multiple links on a page

Technical Architecture

The Trust Engine v2 processes every scan through a deterministic pipeline in which each stage feeds the next. Stages run sequentially where order matters (normalization must precede rules; rules must precede TI because TI results need entity type to know which collectors to call) and in parallel where they are independent (all TI collectors run concurrently).

6.1 Pipeline Stages

Input	Request arrives via web UI, Chrome extension, or REST API. Authenticated via JWT (Supabase Auth) or hashed API key matched against the api_keys table.
Normalize	Entity canonicalized based on scan type. URLs: lowercase host, punycode decode, strip tracking params, drop fragment. Email: split local/domain. Phone: India-first E.164. UPI: extract VPA from intent URI pa= parameter. Text: trimmed.
Daily Limit	Free-tier users (plan='free') checked against daily_scan_count via SECURITY DEFINER RPC. Over-limit throws DailyLimitError (HTTP 429). Pro/Business skip this check.
Rule Engine	Deterministic content, identity, payment, and phone rules run synchronously in under 1ms. No external calls. Produces Signal[] with source='rule_engine'.
Entity Extraction	For text and email inputs, embedded URLs, UPI IDs, phone numbers, and emails are extracted using regex. Each spawns a sub-pipeline (up to 3 URLs, 3 phones).
Redirect Expansion	URL entities are unrolled through redirect chains (up to 10 hops, 2s timeout per hop) before structural and TI analysis. Expands shorteners, catches redirect-based evasion.
Structural Collector	URL structure heuristics: IP host, punycode, brand in subdomain, suspicious TLD, excessive subdomains, @ in path, no TLS, redirect chain length. Zero external calls.
TI Collectors	11 network collectors run concurrently under a 4-second AbortController budget. Each implements appliesTo(), isConfigured(), collect(). Fail-open: timeout returns []. Cache-first: LRU per-source with configurable TTL.
Reputation Engine	Supabase query for community reports on the normalized entity. Result may add: reputation.community_abuse, reputation.prior_scam_verdict, reputation.community_override (hard malicious override), reputation.clean_history (negative signal), reputation.trusted_domain (negative signal).
Risk Engine	Signals accumulated with: effectiveWeight = base_weight x confidence x tier_multiplier (T1=1.0, T2=0.7, T3=0.5). Per-category caps applied. Positive signals summed (capped), negative signals summed, riskScore = positives + negatives, clamped 0-100.
Confidence Calc	coverage = (queried - failed) / queried. reliability = weighted average of source tiers. agreement = 1 - (disagreement ratio). confidence = 0.45*coverage + 0.25*reliability + 0.30*agreement.
Override Check	If any signal carries override='malicious' and confidence >= 0.9: riskScore set to 100. If any signal carries override='allowlist': riskScore capped at 10.

AI Explanation	Positive-weight signals passed to Claude (via OpenRouter) with entity type, risk level, and confidence. Claude returns summary, recommendation, scamType. Never generates numbers. Falls back through model chain on failure.
Persist	Scan, flags, and signals written to Supabase under the user's JWT (RLS). scan_signals table stores the full evidence trail for each scan, enabling verdict reproducibility.
Response	Full ScanResultV2 returned: all fields including engineVersion, processingTimeMs, confidence, signals[], decisionTrace (which sources were queried and which failed).

Scoring Engine -- Signal Reference

Every signal in the trust engine has a documented identifier, category, base weight, source tier, and human-readable label. Weights are versioned alongside the engine. No weight change is allowed without bumping `ENGINE_VERSION`, which is embedded in every scan response. This makes historical verdicts reproducible: given an engine version and the raw input, any verdict can be reconstructed.

7.1 Category Caps

Per-category caps prevent a single category from dominating the score even when multiple signals from the same category fire. This ensures that a URL with many infra signals (shortener + no TLS + punycode + brand in subdomain) does not automatically hit critical risk without supporting evidence from content or reputation.

Category	Cap	Rationale
blocklist	60	Hard TI evidence is strong but should leave room for override signals
reputation	50	Community data is high-signal but may lag new campaigns
content	35	Linguistic patterns alone are insufficient for a high verdict
pay	35	Payment signals are specific but need corroboration for high verdict
identity	30	Identity checks supplement other signals; rarely sufficient alone
infra	30	URL structure signals are easy to fake on both sides
domain_age	30	Newly registered domains are suspicious but not always fraudulent

7.2 Source Tier Multipliers

Tier	Multiplier	Sources	Rationale
T1 (1.0)	100%	Rule engine for hard-coded TRAI rules, GSB, PhishTank, URLHaus, OpenPhish, RDAP, Spamhaus	Highly reliable; structured data or well-maintained feeds
T2 (0.7)	70%	VirusTotal, AbuseIPDB, community reputation, SPF/DMARC checks	Reliable but probabilistic; community-driven or aggregate data
T3 (0.5)	50%	Content/linguistic rules, infra structural heuristics	Indicative patterns; high false-positive rate without corroboration

7.3 Complete Signal Weight Matrix

Signal ID	Category	Wt	Tier	Eff.Wt (T3)	Label
ti.gsb.malware	blocklist	100	T1	100	Google Safe Browsing -- malware
ti.gsb.social_engineering	blocklist	100	T1	100	Google Safe Browsing -- phishing

ti.phishtank.verified	blocklist	100	T1	100	Verified phishing (PhishTank)
ti.urlhaus.malware	blocklist	100	T1	100	Malware URL (URLHaus)
ti.openphish.match	blocklist	100	T1	100	Active phishing (OpenPhish)
reputation.community_override	reputation	100	T1	100	Confirmed by multiple reports
ti.virustotal.ratio_high	blocklist	45	T2	31.5	Flagged by many AV engines
ti.spamhaus.dbl	blocklist	40	T1	40	Domain on Spamhaus blocklist
ti.spamhaus.zen	blocklist	30	T1	30	IP on Spamhaus blocklist
domain.sinkholed	blocklist	30	T1	30	Resolves to sinkhole IP
ti.abuseipdb.high	blocklist	25	T2	17.5	High abuse reports (AbuseIPDB)
ti.virustotal.ratio_low	blocklist	18	T2	12.6	Flagged by some AV engines
ti.abuseipdb.low	blocklist	12	T2	8.4	Some abuse reports (AbuseIPDB)
domain.age_lt_7d	domain_age	30	T1	30	Domain < 7 days old
domain.age_lt_30d	domain_age	25	T1	25	Domain < 30 days old
domain.age_lt_90d	domain_age	15	T1	15	Domain < 90 days old
domain.age_lt_1y	domain_age	6	T1	6	Domain < 1 year old
domain.low_ttl	domain_age	8	T2	5.6	Very short DNS TTL
content.credential_request	content	32	T3	16	Requests OTP/PIN/CVV
content.attachment_or_apk	content	24	T3	12	Install APK / open attachment
content.kyc_request	content	26	T3	13	Asks to update KYC outside app
content.lottery_prize	content	22	T3	11	Prize/lottery bait
phone.premium_rate_prefix	content	35	T1	35	Premium-rate phone prefix
phone.suspicious_sequence	content	20	T2	14	Suspicious digit sequence
content.job_upfront_fee	content	20	T3	10	Job offer with upfront fee
content.too_good_returns	content	20	T3	10	Unrealistic investment returns
content.payment_pressure	content	18	T3	9	Pay now or face consequence
content.urgency_threat	content	16	T3	8	Artificial urgency/threat
content.loan_unsolicited	content	16	T3	8	Unsolicited loan offer
content.contact_offplatform	content	12	T3	6	Move to WhatsApp/Telegram
pay.upi_brand_impersonation	pay	22	T3	11	UPI ID impersonates brand
pay.upi_suspicious_handle	pay	20	T2	14	Scam keyword in UPI handle
pay.upi_intent_name_mismatch	pay	16	T3	8	Payee name vs PSP mismatch

pay.upi_unknown_psp	pay	18	T3	9	Unknown UPI payment provider
pay.collect_request_unsolicited	pay	14	T3	7	Unsolicited collect request
phone.invalid_indian_mobile_format	identity	30	T1	30	Invalid +91 mobile format
identity.sender_domain_mismatch	identity	24	T3	12	Sender domain vs brand mismatch
phone.intl_claiming_indian_bank	identity	20	T2	14	Intl number claiming Indian bank
identity.brand_impersonation	identity	14	T3	7	Brand name in content
identity.free_email_for_company	identity	12	T3	6	Free email for corporate brand
domain.no_mx_yet_emails	identity	8	T2	5.6	No MX record on email sender
infra.punycode_homoglyph	infra	28	T3	14	Lookalike punycode domain
infra.shortener	infra	24	T3	12	Shortened link
infra.host_is_ip	infra	22	T3	11	Raw IP in URL
infra.tls_self_signed	infra	18	T3	9	Self-signed TLS cert
infra.redirect_chain_long	infra	18	T3	9	Long redirect chain
infra.tls_cn_mismatch	infra	16	T3	8	TLS cert hostname mismatch
infra.at_symbol_in_url	infra	16	T3	8	@ symbol in URL path
infra.no_tls	infra	14	T3	7	No HTTPS
infra.suspicious_tld	infra	14	T3	7	Suspicious TLD
infra.brand_in_subdomain	infra	20	T3	10	Brand in subdomain
infra.excessive_subdomains	infra	10	T3	5	Excessive subdomain nesting
infra.via_qr_code	infra	5	T3	2.5	Content from QR code
reputation.community_abuse	reputation	50	T2	35	Reported as scam
reputation.prior_scam_verdict	reputation	30	T2	21	Previously confirmed scam
identity.verified_org	identity	-40	T1	-40	Verified organisation
reputation.trusted_domain	reputation	-30	T1	-30	Trusted established domain
domain.age_gt_5y	domain_age	-20	T1	-20	Domain 5+ years old
identity.spf_dkim_dmarc_pass	identity	-12	T1	-12	SPF/DKIM/DMARC pass
domain.age_gt_2y	domain_age	-10	T1	-10	Domain 2+ years old
reputation.clean_history	reputation	-15	T2	-10.5	No abuse reports

Eff.Wt (T3) column shows effective weight if source tier is T3 (x0.5 multiplier). For T1 signals effective weight equals base weight.

API Reference

The Neural Shield AI REST API is available at the backend endpoint. All requests require an Authorization header. Web users authenticated via Supabase Auth pass a Bearer JWT. Enterprise/developer users pass their API key. Rate limits and daily quotas are enforced per authentication type.

8.1 Authentication

Method	Header	Applies To	Daily Limit
JWT (Supabase)	Authorization: Bearer	Web app, Chrome extension	10 scans/day (Free), Unlimited (Pro/Business)
API Key	Authorization: Bearer	Enterprise integrations	Per-key quota set at onboarding

8.2 Scan Endpoints

Endpoint	Method	Body Fields	Description
POST /api/scan/message	POST	{ "text": "..." }	Analyze a text/SMS/WhatsApp message
POST /api/scan/url	POST	{ "url": "..." }	Analyze a URL (full pipeline, 11 collectors)
POST /api/scan/email	POST	{ "sender", "subject", "body" }	Analyze an email with sender, subject, body
POST /api/scan/phone	POST	{ "phone": "..." }	Analyze a phone number
POST /api/scan/upi	POST	{ "upild": "..." }	Analyze a UPI VPA or intent URI
POST /api/scan/screenshot	POST	multipart/form-data: file	OCR + analyze an uploaded image
POST /api/scan/qr	POST	multipart/form-data: file	Decode QR + analyze payload
POST /api/scan/batch	POST	{ "items": [...] }	Batch analyze up to 10 items (extension)

8.3 Response Format

All endpoints return: { success: true, data: { scamProbability, trustScore, riskLevel, confidence, signals[], flags[], summary, recommendation, scamType, engineVersion, processingTimeMs } }

8.4 Rate Limits

Plan	Requests/min	Daily Scans	Concurrent Requests	SLA
Free	10	10	1	Best effort
Pro	60	Unlimited	5	Best effort
Business	300	Unlimited	20	99.5% uptime

Enterprise	Custom	Custom	Custom	Custom SLA + dedicated infra
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Security and Privacy Architecture

Security is not an afterthought in Neural Shield AI -- it is a constraint that shaped the architecture. A fraud detection platform that stores or leaks user scan data would itself become a target and a liability. Every architectural decision around data storage and access was made with the assumption that the backend may be compromised.

9.1 Row Level Security (RLS)

All user data (scans, scan_flags, scan_signals, audit_logs) is stored in Supabase PostgreSQL with RLS policies enforced at the database layer. The backend never holds a service-role key. Every database write and read on behalf of a user happens under that user's JWT, meaning the database enforces ownership checks independently of application code. A bug in the backend cannot expose one user's scans to another user.

- scans: `user_id = auth.uid()` -- users can only read their own scan history
- scan_flags and scan_signals: joined to scans via scan_id -- transitively protected
- profiles: `user_id = auth.uid()` -- plan and quota data is user-owned
- community_reports: write-only via SECURITY DEFINER RPC -- users can submit reports but not read the raw report table
- api_keys: protected by hashed key matching, never stored in plaintext
- audit_logs: append-only under user JWT -- immutable after write

9.2 No Service-Role Key in Backend

The most consequential security decision in the architecture: the Node.js backend was deliberately built to operate with no Supabase service-role key. All privileged operations (incrementing daily scan counts, writing reputation aggregates, resolving API keys) are implemented as SECURITY DEFINER RPCs -- PostgreSQL functions that run with elevated privileges server-side while the caller needs only the anon key. This means a full compromise of the Node.js process does not grant the attacker access to the entire database.

9.3 Payment Security

Payments are processed through Razorpay via a Supabase Edge Function (Deno runtime). The Razorpay key and secret never touch the Node.js backend. The Edge Function creates a Razorpay order and returns only the order ID to the frontend. The frontend completes the payment UI. On completion, the frontend submits the payment signature to the Edge Function, which verifies the HMAC signature server-side before upgrading the user's plan. At no point does the frontend receive the Razorpay secret, and the backend never sees it.

9.4 API Key Security

Enterprise API keys are hashed (SHA-256) before storage. The raw key is shown to the user exactly once at creation time. Authentication middleware computes the hash of the presented key and compares it

against the stored hash -- the database never contains a usable plaintext key. Daily scan counts for API keys are incremented via the same SECURITY DEFINER RPC pattern used for user quotas.

9.5 Content Privacy

Scanned content (URLs, text, phone numbers, UPI IDs) is stored in Supabase under the user's RLS context. Users can view their own scan history. Scan content is never used for model training. Community reports (which contribute to the reputation engine) are stored in a separate aggregated form -- the individual report data is accessible only via server-side SECURITY DEFINER functions, not directly queryable by users.

Cost and Performance

Cost efficiency is a first-class design goal. A fraud detection platform that costs more than it saves in prevented fraud is not viable as a consumer product. Neural Shield AI's tiered pipeline was designed from day one to minimize API call volume while maximizing detection coverage.

10.1 Cost Per Scan Breakdown

Cost Component	Text Scan	URL Scan (full)	Image Scan
Rules engine	~\$0.000	~\$0.000	~\$0.000
RDAP (domain age)	N/A	~\$0.000 (free)	N/A
Google Safe Browsing	N/A	~\$0.000 (free)	N/A
VirusTotal v3	N/A	~\$0.001	N/A
AbuseIPDB	N/A	~\$0.000	N/A
URLHaus/PhishTank	N/A	~\$0.000 (free feeds)	N/A
OCR (tesseract.js)	N/A	N/A	~\$0.000 (local)
Claude explanation	~\$0.001	~\$0.002	~\$0.002
Supabase (DB ops)	~\$0.000	~\$0.000	~\$0.000
Total estimated	~\$0.001	~\$0.003-0.005	~\$0.015-0.020

10.2 Cache Strategy

- **URL/domain TI results:** Cached in-memory (LRU, max 500 entries) with source-specific TTL. RDAP: 4 hours. GSB/PhishTank/URLHaus: 1 hour. VirusTotal: 2 hours. AbuseIPDB: 1 hour. Spamhaus DNS: 30 minutes.
- **Reputation queries:** Not cached (always fresh from Supabase to reflect newly submitted community reports).
- **Claude explanations:** Not cached per-scan (each explanation is personalized to the specific signal set). Templated fallback explanation is used when AI is unavailable.
- **Cache hit rate:** For popular scam infrastructure (domains/IPs used in mass campaigns), cache hit rates exceed 80% -- dramatically reducing per-scan API costs on high-traffic campaigns.

10.3 Performance Benchmarks

Scan Type	Median Latency	P95 Latency	Breakdown
Text (no embedded URLs)	300-500ms	800ms	Rules (1ms) + Claude explanation (~300ms via OpenRouter)
URL (full TI pipeline)	1500-2500ms	4000ms	Rules + redirect + 11 collectors parallel (budget 4s) + Claude

Phone	200-400ms	700ms	Rules only + Claude (no TI collectors for phone-only)
UPI	200-400ms	700ms	Rules only + Claude
Screenshot (OCR)	1000-2000ms	3500ms	OCR (600-800ms) + text pipeline + Claude
QR code	800-1500ms	3000ms	QR decode + URL or UPI pipeline
Email	1500-2500ms	4000ms	Rules + sender domain TI + Claude

Scalability Architecture

Neural Shield AI is designed to scale from thousands to millions of scans per day without architectural rewrites. The stateless backend, database-enforced RLS, and plug-in collector model were chosen specifically to enable horizontal scaling without coordination complexity.

11.1 Stateless Backend

The Express.js backend runs as Vercel Serverless Functions today -- each request is handled by an independent function instance with no shared in-memory state between instances. Session state is managed via JWT (stateless, validated on each request). The daily scan counter is managed in Supabase via atomic RPC, not in process memory. This means zero-config horizontal scaling: Vercel automatically provisions new function instances under load.

When scan volume outgrows serverless cold-start tolerances, the backend is portable to containerized compute. The Dockerfile exists in the repository. Migration to Google Cloud Run, AWS ECS, or Fly.io requires only updating environment variables and DNS -- no code changes.

11.2 Database Scaling

Supabase PostgreSQL with RLS is the current database layer. Scaling path:

- **Read replicas:** Supabase supports read replicas. The backend's Supabase client can be pointed to replica endpoints for scan history reads without code changes.
- **Connection pooling:** PgBouncer (built into Supabase) handles connection pooling. Serverless function spike traffic is already handled by the pooler.
- **Partitioning:** The scans table can be partitioned by created_at (monthly or quarterly) as volume grows, without changing application query patterns.
- **Migration to managed PostgreSQL:** If volume requires Aurora or Cloud SQL, the Supabase schema is standard PostgreSQL with no Supabase-specific extensions that would prevent migration.

11.3 TI Cache Scaling

The current in-memory LRU cache is per-function-instance on Vercel serverless -- each cold start begins with an empty cache. This is acceptable at current scale but becomes a limitation at high concurrency. The upgrade path requires no code changes to the collector interface:

- **Redis/Upstash:** Replace the in-memory cache adapter with a Redis client. Cache keys are already namespaced by source and entity value. TTL logic is already centralized in the cache module.
- **Edge cache:** For static blocklists (PhishTank, OpenPhish feeds), these can be replicated to Cloudflare KV or Deno KV at the edge, reducing latency to zero for common queries.
- **Distributed bloom filter:** For high-volume URL scanning, a distributed bloom filter can pre-screen URLs against known-bad lists before API calls.

11.4 Adding New Collectors

The collector interface is deliberately minimal. Any new threat intelligence source implements three methods:

appliesTo(entity): boolean -- which entity types this collector handles **isConfigured(): boolean** -- whether required API keys / feeds are set **collect(entity, signal): Promise** -- run the check, return signals[]

The registry automatically includes any registered collector in the parallel execution plan. No changes to the orchestrator, risk engine, or API layer are needed. Adding a new TI source (e.g., TRAI DIP caller ID, CERT-In feed, RBI fraud database) is a self-contained ~80-line implementation.

Market Opportunity

Neural Shield AI operates at the intersection of three major market categories: fraud detection, cybersecurity software, and AI-powered safety tools. Each market is growing rapidly, and Neural Shield AI's multi-modal, consumer-accessible approach is differentiated within all three.

12.1 Market Sizing

Market	2023 Size	2029 Projected	CAGR	Our Addressable Slice
Global Fraud Detection & Prevention	USD 28.8B	USD 63.1B	14.2%	Real-time content screening layer
India Cybersecurity Market	USD 2.0B	USD 3.5B	9.8%	Consumer + SME + fintech segment
Global Identity Verification	USD 11.6B	USD 32.8B	18.9%	Scam entity verification sub-segment
Digital Payments (India UPI)	USD 2.2T vol	Growing	--	Transaction screening and UPI fraud

12.2 TAM / SAM / SOM

Market Tier	Definition	Size	Rationale
TAM (Total Addressable)	All potential buyers of fraud detection software globally	USD 63B (2029)	Every organization and individual exposed to digital fraud
SAM (Serviceable)	Organizations and individuals in India + SE Asia needing multi-modal content screening	USD 4-6B	Fintechs, telecoms, e-commerce, consumers in high-fraud markets
SOM (Serviceable Obtainable)	Neural Shield AI's realistic 3-year addressable segment	USD 40-80M ARR	Based on 500K pro users + 200 B2B API customers by Year 3

12.3 Target Customer Segments

- **Individual consumers (B2C, freemium):** Urban and semi-urban smartphone users in India (600M+ active internet users). Primary acquisition via web app and Chrome extension. Converts to Pro via daily limit friction. LTV primarily from subscription revenue.
- **Fintech companies:** Digital lenders, UPI apps, neobanks, BNPL platforms. Need to screen UPI IDs, URLs in loan applications, and identify synthetic fraud signals. API integration; pricing per scan volume. High willingness to pay due to direct fraud cost reduction.
- **Telecom operators:** Bharti Airtel, Jio, BSNL. TRAI DIP mandate requires carriers to screen commercial SMS content. Neural Shield AI's text pipeline + phone rules can integrate at the SMS gateway level. Very high volume; require dedicated infrastructure and SLA.
- **E-commerce platforms:** Meesho, ONDC network participants, Instagram commerce. Seller fraud (fake payment screenshots as delivery proof), buyer fraud (chargeback after QR payment). Need screenshot OCR and UPI verification at transaction layer.

- **Banks and NBFCs:** Customer-facing scam alert integration -- screen URLs from customer support chat, verify UPI IDs before transfer, flag suspicious incoming collect requests. Compliance and customer protection angles.
- **Government and regulatory bodies:** CERT-In, TRAI Consumer Affairs, MeitY, RBI Ombudsman digital interface. Bulk threat intelligence, incident aggregation, and reporting dashboards. Procurement-driven; longer sales cycle but high contract values.
- **Elderly-care and NGO programs:** White-label deployment through NGOs working with elderly populations, banks' senior citizen programs, or Doordarshan awareness campaigns. Simplified UI, vernacular language support.

Business Model and Pricing

Neural Shield AI operates a freemium model for consumer channels (web + extension) and a subscription/usage model for API access. The freemium layer drives user acquisition and organic growth; the API layer drives revenue. Enterprise licensing provides high-value contracts for telecom and government customers.

13.1 Pricing Tiers

Tier	Price	Daily Scans	API Access	Target User
Free	INR 0 / USD 0	10/day	No	Casual users, evaluation
Pro	INR 299/mo / USD 3.6/mo	Unlimited	No	Power users, professionals
Business	INR 999/mo / USD 12/mo	Unlimited	300 req/min	SMEs, developers
Enterprise	Custom	Custom quota	Custom SLA	Banks, telecoms, government

13.2 Revenue Streams

- **Subscription revenue (Pro/Business):** Recurring monthly or annual subscription. High margin -- cost per unlimited scan for a typical Pro user is under USD 0.50/month at current usage patterns.
- **API usage revenue (Business/Enterprise):** Per-scan pricing above free tier allowance, or flat monthly fee for dedicated rate limits. B2B contracts provide predictable ARR.
- **Enterprise licensing:** White-label deployments for banks or telecom carriers. Annual license fee covering infrastructure, SLA, and support. One-time integration cost plus annual renewal.
- **Data intelligence (future):** Aggregated, anonymized threat intelligence reports for regulatory bodies. Trend data on scam campaign patterns by geography, modality, and brand targeted.

13.3 Unit Economics

Metric	Free User	Pro User	Business API Customer
Monthly Revenue	INR 0	INR 299	INR 999+
Avg Scans/Month	~50	~500	~10,000
Cost of Goods (AI+TI)	~INR 2	~INR 30	~INR 600
Infrastructure Share	~INR 2	~INR 10	~INR 50
Gross Margin	Negative	~INR 259 (87%)	~INR 349 (35%)
LTV (24-month est.)	INR 0	INR 5,000+	INR 20,000+

Competitive Analysis

The fraud detection landscape is fragmented. No direct competitor offers a unified multi-modal pipeline at the price point and accessibility level that Neural Shield AI targets. Competitors either serve only enterprises (too expensive), handle only one modality, or use black-box AI without auditable scoring.

14.1 Competitive Landscape

Category	Key Players	Limitations vs Neural Shield AI
Enterprise Fraud Platforms	NICE Actimize, LexisNexis, BioCatch	Priced for banks (USD 100K+/year); no consumer-facing channel; no multi-modal content analysis
URL/Domain Scanners	VirusTotal, URLScan.io, Sucuri	URL-only; no text, UPI, phone, image; no AI explanation; no community reputation
Email Security	Mimecast, Proofpoint, Barracuda	Enterprise email gateways; no consumer channel; no UPI/phone/QR analysis
India-specific Apps	Various telco-branded apps	Phone number lookup only; no URL/UPI/image analysis; carrier data only; no AI
AI Chatbots (DIY)	ChatGPT, Claude (direct)	No structured output; inconsistent scoring; no TI integration; no community data; no API
Browser Extension Security	MetaMask Snaps, DappRadar	Web3/crypto focused; no Indian payment context; no UPI/phone/image analysis

14.2 Feature Comparison

Feature	Neural Shield AI	VirusTotal	Enterprise Platforms	DIY AI Chatbot
Text / SMS analysis	YES	No	Partial	Partial
URL scanning (11 collectors)	YES	YES	YES	No
UPI / Payment ID analysis	YES	No	No	No
Phone number analysis	YES	No	Partial	No
Screenshot OCR analysis	YES	No	No	No
QR code decoding	YES	No	No	No
Chrome extension	YES	YES	No	No
Community reputation	YES	YES	Partial	No
Auditable deterministic score	YES	No	No	No
Consumer price point	YES	YES	No	YES
REST API	YES	YES	YES	No

Indian regulatory context	YES	No	No	No
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14.3 Competitive Moat

- **Signal weight matrix:** The scoring matrix has taken months to calibrate. Getting the weights right for Indian fraud patterns (TRAI prefix tables, UPI PSP lists, brand name sets) is not a 30-day project for a new entrant.
- **Community reputation data:** Community report aggregates improve with every submitted report. This is a compounding data moat -- early users generate better data for future users.
- **Multi-modal architecture:** A competitor who builds only URL scanning has to rebuild from scratch to add UPI analysis. Neural Shield AI's entity-agnostic scoring model means new modalities plug in at the collector level.
- **Indian payment context:** TRAI prefix tables, NPCI PSP lists, Indian brand canonical domains -- this domain knowledge takes time to compile and maintain. It is not available in any open dataset.
- **Regulatory alignment:** CERT-In and TRAI are mandating cyber fraud reporting infrastructure. Neural Shield AI's architecture (deterministic scores, full signal trails, RLS-protected data) is designed for regulatory audit requirements.

Go-to-Market Strategy

Neural Shield AI's GTM strategy is phased. Phase 1 establishes organic user acquisition through the free product and early API access. Phase 2 uses validated product-market fit to pursue fintech and telecom partnerships. Phase 3 targets government and regulatory contracts with a proven track record.

Phase 1 -- Organic Consumer Acquisition (Months 1-12)

- **Distribution:** Web app at neuralshield.ai, Chrome extension on the Web Store, social media presence targeting digital safety awareness.
- **Content:** Public 'scam of the week' posts analyzing real-world scam messages (with PII removed) -- builds credibility and demonstrates the platform's capability.
- **Developer community:** Free API tier for developers; public documentation and Postman collection. Developer integrations create organic B2B inbound.
- **Referral:** Share scan results on WhatsApp / social media with a branded card -- 'Scanned by Neural Shield AI' with the verdict. Viral acquisition from forwarded scam messages.
- **Target KPIs:** 50,000 registered users, 500 Pro subscribers, 50 Business API customers by Month 12.

Phase 2 -- Fintech and Telecom Partnerships (Months 12-24)

- **Fintech integration:** Direct outreach to UPI apps, lending platforms, and BNPL providers. Use Phase 1 scan volume data to demonstrate ROI on fraud reduction.
- **Telecom carrier pilots:** Partner with one carrier for SMS content screening pilot. TRAI DIP compliance angle creates regulatory pull.
- **Bank consumer protection:** White-label the consumer-facing app for a bank's senior citizen program or fraud alert notification system.
- **Target KPIs:** 3 fintech API customers, 1 telecom pilot, 200,000 registered users, INR 1Cr ARR by Month 24.

Phase 3 -- Enterprise and Government (Months 24-36)

- **CERT-In / MeitY:** Bid for national cyber fraud intelligence platform. Use Phase 2 data on scam campaign patterns as the proposal's evidence base.
- **RBI Ombudsman integration:** Fraud reporting API that feeds into the RBI's digital complaint system.
- **Bank fraud desk partnerships:** API integration with bank fraud desks -- when a customer calls to report a scam, the agent uses Neural Shield AI to verify the reported content in real time.
- **Target KPIs:** 1 government contract, 2 bank partnerships, INR 5Cr+ ARR by Month 36.

Financial Projections (3-Year)

The following projections are based on conservative assumptions: free-to-paid conversion rate of 3% (industry benchmark for B2C SaaS is 2-5%), average Pro revenue of INR 299/month, and B2B API customer acquisition of 5 new customers per month from Month 12. All figures in Indian Rupees.

16.1 User and Revenue Growth

Metric	Year 1	Year 2	Year 3
Registered Users	50,000	200,000	750,000
Pro Subscribers	500	4,000	15,000
Business API Customers	50	200	500
Enterprise Contracts	0	2	5
Monthly Pro Revenue (INR)	1,49,500	11,96,000	44,85,000
Monthly API Revenue (INR)	49,950	1,99,800	4,99,500
Monthly Enterprise (INR)	0	5,00,000	12,50,000
Total Monthly Revenue (INR)	~2L	~19L	~62L
Annual Revenue (INR)	~24L	~2.3Cr	~7.4Cr

16.2 Cost Structure

Cost Category	Year 1 (Monthly)	Year 2 (Monthly)	Year 3 (Monthly)
Infrastructure (Vercel + Supabase)	INR 15,000	INR 40,000	INR 1,20,000
AI / LLM (OpenRouter)	INR 20,000	INR 80,000	INR 2,50,000
TI API calls (VT, ABDB, etc.)	INR 10,000	INR 35,000	INR 90,000
Team / Salaries	INR 2,00,000	INR 8,00,000	INR 25,00,000
Marketing / Acquisition	INR 30,000	INR 1,50,000	INR 5,00,000
Legal / Compliance / Misc.	INR 20,000	INR 50,000	INR 1,00,000
Total Monthly Opex	~INR 2.95L	~INR 10.55L	~INR 34.6L

16.3 Path to Profitability

Year 1 operates at a modest deficit funded by founder capital and early investor support. Year 2 sees revenue accelerate past break-even at Month 18-20 driven by B2B API growth. Year 3 achieves sustained profitability with enterprise contracts providing predictable ARR. Gross margins are high (85%+ on software subscriptions) and improve with scale as TI API costs are amortized over more scans.

Product Roadmap

The roadmap is organized by capability domain and time horizon. All dates are targets, not commitments. Technical feasibility has been validated for Q3-Q4 2025 items. Q1-Q2 2026 items are architecture-validated but require dedicated engineering time.

Q3 2025 (Current Quarter)

Feature	Status	Notes
Voice / call scam detection pipeline	In development	Real-time transcription + LLM analysis of audio
Hindi keyword set	In progress	50+ scam phrase patterns in Devanagari script
TRAI DIP integration	Scoping	Caller ID enrichment for phone scanner
Android SDK (beta)	Design phase	In-app scam detection for third-party apps
Enterprise batch API GA	Testing	SLA-backed high-volume endpoint

Q4 2025

Feature	Status	Notes
Tamil and Bengali keyword sets	Planned	Additive to rules engine, no architecture change
Deepfake image detection (v1)	Planned	Detect AI-generated/manipulated images in scam context
Carrier SMS screening API	Planned	High-volume endpoint for telecom carrier gateway integration
Regulatory reporting dashboard	Planned	Aggregate scam pattern reports for CERT-In / TRAI
White-label product for banks	Planned	Branded consumer app for bank's anti-fraud program

2026 and Beyond

- **Financial document fraud detection:** Analyze PDF bank statements and KYC documents for fabrication patterns -- addresses the growing market of loan application fraud using manipulated documents.
- **Video and audio deepfake detection:** Extend the image pipeline to short video clips and audio recordings. Critical for detecting AI-synthesized voice calls impersonating bank officials.
- **Real-time browser integration (v2):** Extension v2 with page-level scanning -- analyze all links, forms, and payment requests on a page without user-initiated action.
- **Multi-jurisdiction expansion:** South East Asia (Indonesia, Philippines, Vietnam) have similar UPI-equivalent payment fraud patterns. The engine's modular design supports adding new PSP lists and brand sets per country.
- **CERT-In threat sharing integration:** Bidirectional API with CERT-In's ISAC -- Neural Shield AI receives government-sourced threat intel and contributes aggregated community report data.
- **Federated reputation network:** Partner with other fraud detection platforms to share anonymized community report hashes -- expanding the reputation database without sharing raw user data.

Team and Vision

Neural Shield AI was founded by a technologist with deep experience in full-stack product development, AI integration, and consumer-facing security tooling. The founding vision was shaped by observing first-hand how people in Tier 2 and Tier 3 cities in India -- including family members -- were defrauded by increasingly sophisticated digital scams that existing tools had no answer for.

18.1 Founder Background

- Full-stack engineering background across multiple production systems
- Experience building AI-augmented products using large language models in production
- Deep understanding of the Indian fintech and UPI payment ecosystem
- Hands-on builder -- the entire current tech stack (Trust Engine, Chrome extension, web app, backend API) was built by the founding team

18.2 Hiring Plan

Role	Timeline	Priority	Rationale
Senior Backend Engineer	Q3 2025	Critical	Scale TI pipeline; voice detection engineering
ML/NLP Engineer	Q3 2025	High	Vernacular models; deepfake detection research
Product Designer	Q4 2025	High	Enterprise dashboard UX; white-label product
Enterprise Sales	Q4 2025	High	Telecom and government pipeline development
Data Science / Analyst	Q1 2026	Medium	Scam campaign analytics; regulatory reports
DevSecOps Engineer	Q1 2026	Medium	SOC 2 compliance; penetration testing; CI/CD

18.3 Vision -- Where We Are Going

The ten-year vision for Neural Shield AI is to become the trust layer for digital interactions in emerging markets -- the equivalent of HTTPS for web security, but for the content that travels over it. Just as HTTPS tells you the connection is secure, Neural Shield AI tells you the content is trustworthy. We want to reach a state where any digital platform -- a messaging app, a payments gateway, a government portal -- can embed Neural Shield AI verification with a single API call, and any individual can verify any piece of content they receive in under two seconds.

The scam problem will get worse before it gets better. Generative AI is lowering the cost of personalized phishing messages, synthetic voice calls, and believable fake documents. The window to build the infrastructure that can detect these threats at scale is now, before the attack techniques become too sophisticated to catch with rule-based systems alone. Neural Shield AI's hybrid architecture -- deterministic rules + AI explanation -- is designed for this trajectory: rules catch the known patterns at zero marginal cost, and the AI layer adapts as new patterns emerge.

Risk Factors and Mitigations

Risk	Likelihood	Impact	Mitigation
TI API dependency (VirusTotal, GSB outage)	Medium	Medium	All collectors fail-open. Confidence drops but scan completes. Multiple redundant sources per threat category.
AI model cost increase (OpenRouter pricing)	Low	Medium	Claude called only for explanation (~300 tokens). Model fallback chain to cheaper models. Templated explanation as zero-cost fallback.
New scam vector not covered by current rules	High	Low-Medium	Community reports allow fast crowd-sourced flagging. New collector or rule can be deployed without architecture change within days.
Regulatory change (data localization, AI Act equivalent in India)	Medium	High	Data stored in Supabase (can be geo-restricted). No model training on user data. RLS-enforced data isolation simplifies compliance audits.
Competitor with similar product from a large player (Google, Meta)	Low-Medium	High	Indian regulatory context and community data are moat. Partner with large players rather than compete. B2B moat is stickier than consumer.
False positive rate causing user distrust	Medium	High	Auditable scoring with visible signals. Users can understand WHY a verdict was given. Confidence score communicates uncertainty explicitly.
Supabase platform risk (pricing change, acquisition)	Low	Medium	Standard PostgreSQL schema. Migration to managed RDS or Cloud SQL is a 2-week engineering effort. No Supabase-proprietary extensions in schema.

Appendix -- Technical Glossary

Term	Definition
Trust Engine v2	The deterministic scoring pipeline introduced in Week 2 of the build. Distinguishes Neural Shield AI from AI-only detection.
Signal	A typed data structure with: id (string), category, weight (signed), confidence (0-1), source, sourceTier (1-3), optional override, optional evidence.
effectiveWeight	$\text{signal.weight} \times \text{signal.confidence} \times \text{TIER_MULTIPLIER}[\text{signal.sourceTier}]$. The actual contribution of a signal to the risk score.
Category cap	Maximum total effective weight contribution from a single signal category. Prevents any one source type from dominating the score.
Override: malicious	A special signal flag that, when present with confidence ≥ 0.9 , forces riskScore = 100 regardless of other signals.
Override: allowlist	A special signal flag that caps riskScore at 10 regardless of positive signals. Used for verified organizations.
Fail-open	Collector design principle: any error (timeout, API failure, parse error) returns an empty signal array and marks the source as 'failed'. The scan never crashes.
RDAP	Registration Data Access Protocol. The successor to WHOIS for domain registration data. Used to determine domain age.
PSP	Payment Service Provider. In UPI context: the bank or fintech that provides the UPI handle (e.g. oksbi = SBI via PhonePe).
VPA	Virtual Payment Address. The UPI identifier in handle@psp format (e.g. pranj@oksbi).
TRAI DIP	Telecom Regulatory Authority of India - Distributed Ledger Technology for Commercial Communication. India's SMS sender ID and template registration system.
ENGINE_VERSION	Semantic version string embedded in every scan response. Format: trust-engine@X.Y.Z. Allows exact verdict reproduction for audit purposes.
RLS	Row Level Security. PostgreSQL feature enforcing per-row access policies. Ensures users can only access their own data at the database level.
SECURITY DEFINER	PostgreSQL function attribute that executes with the privileges of the function owner, not the caller. Used for privileged operations without a service-role key.
E.164	International telephone numbering format: +[country code][number]. Example: +919876543210. Neural Shield AI normalizes all phone inputs to E.164.
Confidence score	Composite metric (0-1): 45% signal coverage (fraction of configured sources that returned results), 25% source tier reliability, 30% signal agreement.

Sample Scan Analyses -- Real-World Examples

The following examples illustrate how the trust engine processes real-world scam content. Each example shows the input, fired signals, effective weights, and final verdict. These are representative of actual scam patterns encountered in the Indian digital ecosystem.

Example 1: KYC Phishing SMS

Input (Text scan): "Dear SBI customer, your account will be blocked due to KYC expiry. Click <http://sbi-kyc-update.xyz> to update immediately. Ignore at your own risk. -SBI Team"

Fired Signal	Source	Eff. Weight	Evidence	
content.kyc_request	rule_engine	13.0	Keyword: KYC	
content.urgency_threat	rule_engine	8.0	Keywords: blocked, immediately	
identity.brand_impersonation	rule_engine	7.0	Brand: SBI	
domain.age_lt_7d	rdap	30.0	Domain registered 2 days ago	
identity.sender_domain_mismatch	rule_engine	12.0	Claimed SBI; domain sbi-kyc-update.xyz	
infra.suspicious_tld	structural	7.0	TLD: .xyz	
ti.urlhaus.malware	urlhaus	100.0	OVERRIDE: malicious	
Verdict	Risk Score	Trust Score	Confidence	Scam Type
CRITICAL (override)	100	0	0.92	kyc_phishing

Example 2: Fake Prize UPI Collect Request

Input (UPI scan): prize.winner2025@hdfcbank Context: "Congratulations! You have won INR 50,000 in KBC Lucky Draw. Accept the collect request to receive your prize."

Fired Signal	Source	Eff. Weight	Evidence	
pay.upi_suspicious_handle	rule_engine	14.0	Keyword in handle: prize, winner	
content.lottery_prize	rule_engine	11.0	Keywords: won, prize, lucky draw, KBC	
pay.collect_request_unsolicited	rule_engine	7.0	Context: collect request, accept	
identity.brand_impersonation	rule_engine	7.0	Brand: KBC mentioned in context	
Verdict	Risk Score	Trust Score	Confidence	Scam Type
HIGH	39.0	61	0.71	lottery_prize_fraud

Note: PSP 'hdfcbank' is on the KNOWN_PSP allowlist, so no unknown_psp signal fires. The handle and context signals are sufficient for a HIGH verdict.

Example 3: Investment Fraud WhatsApp Message

Input (Text scan): "Join our exclusive Telegram trading group. Guaranteed 15% daily returns on crypto investments. Risk-free strategy used by top traders. DM for access. Limited spots. Contact: +1-800-INVEST-NOW"

Fired Signal		Source	Eff. Weight	Evidence
content.too_good_returns		rule_engine	10.0	15% daily returns, guaranteed, risk-free
content.contact_offplatform		rule_engine	6.0	Telegram group
phone.intl_claiming_indian_bank		rule_engine	14.0	Non-Indian number in investment context
content.urgency_threat		rule_engine	8.0	Limited spots
Verdict	Risk Score	Trust Score	Confidence	Scam Type
MEDIUM	38.0	62	0.65	investment_fraud

Example 4: Legitimate Bank Communication (True Negative)

Input (Email scan): From: alerts@hdfcbank.com Subject: Your HDFC Bank statement for May 2025
Body: "Dear customer, your e-statement for savings account XX1234 is ready. View it at <https://netbanking.hdfcbank.com/netbanking>"

Fired Signal		Source	Eff. Weight	Evidence
identity.brand_impersonation		+7.0	rule_engine	Brand: HDFC mentioned
domain.age_gt_5y		-20.0	rdap	hdfcbank.com registered 2001
identity.spf_dkim_dmarc_pass		-12.0	spf_check	All email auth headers pass
reputation.trusted_domain		-30.0	reputation_db	hdfcbank.com in trusted list
Verdict	Risk Score	Trust Score	Confidence	Scam Type
SAFE	0 (capped at 0)	100	0.91	none

Negative signals bring the score below zero; it is clamped to 0. This is the correct behaviour -- well-established domains with valid email authentication should never show as risky.

Frequently Asked Questions

Technical Questions

Q: How does the engine handle a URL it has never seen before?

A: RDAP gives a domain age verdict regardless of whether the URL has been seen before -- a domain registered yesterday is flagged even on its first scan. Structural heuristics (suspicious TLD, IP in host, brand in subdomain) fire on any URL. Combined with content signals from the surrounding message, the engine can produce a meaningful verdict for a brand-new URL. The confidence score will be lower (reflecting lower TI coverage) but the risk level is still informative.

Q: What happens when all TI collectors fail (network outage)?

A: All collectors fail-open. If every network collector times out, the scan still completes using rule engine signals, structural heuristics, and reputation data. The confidence score drops significantly (coverage component = 0), which is reflected in the response. The verdict is marked as low-confidence but the scan does not return an error.

Q: Can the engine be fooled by a scammer who learns the rules?

A: Partly. A scammer who knows that 'kyc' in a UPI handle triggers a flag could avoid that keyword. But the signals are cumulative -- removing one signal rarely drops a verdict from High to Safe. The combination of domain age, brand impersonation, urgency language, and suspicious UPI structure is harder to evade entirely. Additionally, the community reputation layer catches repeat offenders before their content changes.

Q: How is the confidence score different from the scam probability?

A: scamProbability is derived from the risk score -- it reflects how many scam signals fired and how strong they were. confidence reflects how much signal coverage supported the verdict -- whether we had the opportunity to check many sources, not just whether those sources found something. A high-confidence Safe verdict means we checked everything and found nothing. A low-confidence Safe verdict means few sources were available to check.

Q: Why does the engine use an LLM only for explanation, not scoring?

A: Reproducibility and auditability. An LLM scoring approach produces different verdicts for the same input on different runs (temperature-dependent). It cannot cite which specific signals contributed to the score. It cannot be audited or tested with a fixed test suite. The deterministic rules + TI pipeline produces identical results for identical inputs with the same ENGINE_VERSION. The LLM's role is to translate that structured verdict into language a non-technical user can act on.

Business Questions

Q: How do you prevent misuse of the API by bad actors checking whether their scam will be detected?

A: API access requires authentication. Free tier allows only 10 scans per day. Pro and Business tiers require a verified payment method. Scan history is logged under the user's RLS context. High-volume unusual patterns trigger rate limiting. The risk of adversarial probing exists in any detection system -- the mitigation is rate limiting, authentication, and the fact that the deterministic signal matrix is versioned and updatable.

Q: What is the data retention policy?

A: Scan records are retained indefinitely in Supabase under the user's RLS context -- users can view and delete their own history. Aggregated community report data (not linked to individual users) is retained indefinitely as it contributes to the reputation engine. Users can request full account deletion which removes all their scan history.

Q: How quickly can new scam patterns be added to the engine?

A: New content rules (regex patterns) can be added and deployed within hours -- they are data in the WEIGHTS matrix, not code. New TI collectors require ~80 lines of TypeScript and a deployment. Community reports can flag new patterns within minutes of the first report -- the reputation engine is always live. A coordinated response to a new scam campaign can be live within 24 hours.

Q: Is the platform compliant with India's DPDP Act?

A: The architecture is designed for compliance. User data is stored under RLS with user consent at account creation. No data is shared with third parties except the TI APIs (which receive entity values like URLs or IPs, not user PII). The platform does not train models on user scan data. Formal DPDP compliance certification is planned for Q1 2026 with legal counsel.

Investment Ask and Use of Funds

Neural Shield AI is raising a pre-seed round to fund the transition from a founder-built MVP to a team-led, revenue-generating business. The current product is production-ready and generating early revenue. The raise is to accelerate the roadmap items that require dedicated engineering (voice detection, vernacular languages, carrier integration) and to hire the enterprise sales capacity needed to close B2B contracts.

23.1 Funding Ask

Round	Ask	Instrument	Timeline
Pre-seed	USD 500K - USD 750K (INR 4-6 Cr)	SAFE or Convertible Note	Q3 2025

23.2 Use of Funds (18-Month Plan)

Category	Allocation	Amount (INR)	Key Outcomes
Engineering (2 engineers, 18 months)	45%	~1.8 - 2.7 Cr	Voice pipeline, vernacular, deepfake v1, batch API GA
Sales and Business Development	20%	~80L - 1.2 Cr	Enterprise sales hire; fintech and telecom outreach
Marketing and User Acquisition	15%	~60L - 90L	Performance marketing, content, developer community
Infrastructure and TI APIs (18 months)	10%	~40L - 60L	Vercel, Supabase, VirusTotal, AbuseIPDB at scale
Legal, Compliance, and Admin	10%	~40L - 60L	DPDP compliance, IP filing, company setup

23.3 Milestones Funded by This Round

- Voice scam detection pipeline: live in production with Hindi support (Month 6)
- Tamil and Bengali language keyword sets deployed (Month 8)
- Enterprise batch API with SLA guarantees: GA (Month 6)
- First telecom carrier pilot signed (Month 9)
- First bank white-label partnership signed (Month 12)
- 200,000 registered users, INR 1 Cr ARR (Month 18)
- Series A readiness: unit economics proven, enterprise pipeline built (Month 18)

Closing Statement

Neural Shield AI is being built because the threat is real, growing, and systematically underserved. The platform is not a research prototype -- it is a production system processing real scans for real users today. The trust engine is versioned, auditable, and tested. The Chrome extension is live. The REST API accepts enterprise traffic. The community reputation layer is active. The work to reach this point took months of careful engineering: calibrating signal weights, building fail-open collector infrastructure, wiring threat intelligence APIs, and ensuring that the scoring pipeline is reproducible to the digit. The next phase -- voice detection, vernacular language support, carrier integration, government contracts -- is an engineering and distribution challenge, not an architecture one. The foundation is solid. The moat is being built with every scan, every community report, and every new collector added. The goal is a world where anyone, regardless of digital literacy or technical sophistication, can verify before they trust -- in under two seconds, in any language, on any channel. That world is closer than it looks.

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